# A Cross National Network Analysis of Social Political Attitudes from the European Social Survey

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## 1. Introduction

In social and political sciences, one of the challenges is to define and to understand how political identification is formed. In this study, we attempt to answer this question using a network approach based on survey data from eight European countries (Czech Republic, Finland, France, Great Britain, Greece, Hungary, Italy, Netherlands). As nodes in this network, we used social attitude markers - trust in politicians, trust in police, trust in scientists, trust in the legal system, left-right self-placement, religiousness, state interference for economic redistribution, attitudes toward immigrants, and personal responsibility for climate change. The network approach allows us to map correlations between attitudinal variables without imposing a priori structure.

One of the first ideas to reduce these markers to a single organising principle is the left-right scale. However, for cross-national research it has serious limitations, because the idea of "left" and "right" differs across countries. A similar problem already appeared in Adorno's *The Authoritarian Personality* (1950), where the F-scale was designed in such a way that authoritarian answers always aligned with right-wing positions and effectively excluded left-wing authoritarianism from measurement altogether - therefore the desire to strictly limit defining paradigm can lead to methodological trap. To avoid this kind of bias, we drew on Duckitt's (2001) dual-process model, which distinguishes between two psychological orientations: right-wing authoritarianism (RWA) and social dominance orientation (SDO). Unlike the left-right scale, these constructs are not attached to specific political parties or national contexts: RWA captures attitudes toward authority, conformity, and social order; SDO captures preferences for group-based hierarchy and inequality.

The network approach has been applied to political attitudes before (Dalege et al., 2017), but no study to date has used it to test whether RWA and SDO emerge spontaneously from attitudinal data collected independently of these constructs, nor whether their structural role is consistent across different national contexts.

Drawing on Duckitt's (2001) dual-process model, we ask: **do RWA and SDO emerge as organising forces from attitudinal data collected independently of these constructs, and is their structural role consistent across eight European countries or shaped by national context**?

To answer this question, we estimate a network of partial associations using a polychoric correlation model, specifically designed for ordinal survey data. This allows us to visualise how attitudes cluster together and to compare these structures across eight European countries.

## 2. Problem and Motivation

The central problem this study addresses is the relationship between sociopolitical attitudes and political identity is interconnected, and whether that relationship follows a structured social logic or is shaped primarily by a cultural context. Theoretically, Duckitt's dual-process model proposes that two attitudinal orientations which we meant before - right-wing authoritarianism (RWA) and social dominance orientation (SDO) - serve as an intermediate layer between underlying values and observable social behaviour and outcomes. If this social attitude model holds across contexts, we would expect the same attitudinal clusters to appear regardless of country. Practically, understanding whether ideological structure is universal or culturally contingent matters for how we interpret political polarisation, trust in institutions, and outgroup attitudes in modern Europe.

The most direct way to measure RWA and SDO is to use pre-designed scales, such as Altemeyer's (1981) RWA scale or Pratto and colleagues' (2013) SDO scale. These scales have been validated extensively and remain the standard in the field. However, using them risks circularity: if we ask questions explicitly designed to tap RWA and SDO, we cannot test whether these dimensions emerge spontaneously from the broader attitudinal space. A stronger test of Duckitt's dual-process model is to ask whether RWA and SDO arise as organising forces from data collected independently of these constructs.

We therefore take a different approach: we do not administer RWA or SDO scales ourselves, we use existing data from the European Social Survey (ESS) collected in 2021-2022 and select attitude markers that, theoretically, should relate to these two dimensions: trust in institutions (police, scientists, legal system, politicians), left-right self-placement, religiousness, redistribution preferences, attitude toward immigrants and feeling of responsibility for climate change. If Duckitt's model is robust, RWA and SDO should emerge as latent organising principles even from these indirect indicators. If they do not, the model's universality may be questioned.

To test whether the structure is universal or culturally contingent, we use a network approach. We formulate a null hypothesis ($H_0$) that the network structures of different countries are statistically equivalent (i.e., no difference in edge weights or global connectivity). We then apply a permutation-based Network Comparison Test (NCT) to assess whether we can reject $H_0$ in favour of country-specific differences. To answer a research question we also estimate a network of partial associations using *a polychoric correlation model*, which is specifically designed for ordinal survey data. This allows us to visualise how attitudes cluster together and to compare these clusters across countries.

Each attitude marker is treated as a node, and we estimate the strength of associations between them. The data are ordinal (Likert-type scales items from ESS), so we use *polychoric correlation*, which treats ordinal items as imperfect indicators of underlying continuous latent variables. This method is specifically designed for survey data and produces fewer false positives than Pearson-based methods. For network estimation, we use *EBICglasso* (Extended Bayesian Information Criterion graphical lasso), which applies regularisation to produce sparse networks while being more stable than standard GLasso for psychometric data since we found out empirically that the GLasso method was not accurate in case of missing values or lack of data. This gives us, for each of the eight countries, a visualised network of partial associations between attitude markers: positive edges (green) and negative edges (red), node size reflecting weighted degree (centrality), and node colour indicating modularity class (clusters of nodes that are more connected to each other than to the rest of the network). Each country-level network is undirected, weighted, and monomodal: nodes represent individual attitude markers, and edges represent regularised partial correlations between them.

We do not claim that the networks we estimate reveal the one true structure. Rather, we treat them as one possible representation, shaped by our theoretical and methodological choices. Our contribution is empirical: we test whether Duckitt's dual-process model holds when RWA and SDO are not directly measured but inferred from the network structure of related attitudes across eight European countries.

## 3. Datasets

The data used for this study relies on the European Social Survey (ESS), a cross-national survey oriented to monitoring changes in values and perspectives of European citizens (ESS ERIC, 2025). The ESS provides open data under the CC BY-SA 4.0 license to support accessibility for research and was published on December 9, 2025. In particular, we focus on the version *ESS round 10 - 2020. Democracy, Digital social contacts*[1] which includes various questions related to topics such as politics, socio-economic status, social values, social exclusion, discrimination, religion, democracy, demography and more, across 31 countries in Europe. Depending on the country, the data was gathered between 2020 and 2022 either by self-completion or (online) face-to-face interviews due to COVID constraints.[2] To establish a common methodological ground and acknowledge the limitations of this study's possible scope, this psychometric network analysis uses data from countries that followed the standard interview procedure.

Drawing on this dataset, eight countries were selected to reflect the geographic and ideological diversity of Europe: Czech Republic (CZ), Finland (FI), France (FR), Great Britain (GB), Greece (GR), Hungary (HU), Italy (IT), and the Netherlands (NL). The sample size for each country never falls below 1000 respondents, guaranteeing a statistically significant group and allowing for consistent data collection that mirrors the population's attitude.

Regarding the demographics of the case groups, respondents were chosen following two criteria: to be aged 15 or older to be resident in a private household in any of the countries selected. The sample was constructed through a random address-based selection[3], regardless of nationality, citizenship or legal status (ESS, 2025)[4].

The analysis centres on nine variables, grouped into two theoretically motivated clusters and reports on self-perceived ideology and deference to authority. The *Trust cluster* reports on four institutional actors: the legal system (*trstlgl*), the police (*trstplc*), scientists (*trstsci*), and politicians (*trstplt*). The *Ideological cluster* captures political self-placement (*lrscale*), state interference with income redistribution (*gindif*), religiosity (*rlgdgr*), perceptions of immigration (*imwbcnt*), and personal responsibility for climate change (*ccrdprs*). The survey consists of closed ordinal answers where respondents position their views on Likert-type scales (0-10). Table 1 provides the exact wording and ordinal scales for each variable.

[1] Dataset available here [https://doi.org/10.21338/ess10e03_3]

[2] «ESS data collection is based on an hour-long face-to-face interview. [...] a total of 9 countries switched to a self-completion approach, while 22 countries used ESS's usual face-to-face fieldwork approach. [...] Caution is advised when making comparisons between countries using face-to-face and self-completion approaches at ESS.» (ESS ERIC, 2025).

[3] «[...] the participating countries use an address-based sample, meaning that one person is selected at random at each address in a random sample of addresses.» (Kaminska, 2020)

[4] «All persons aged 15 and over resident within private households, regardless of their nationality, citizenship, language or legal status, in the countries as listed in the "Geographical Unit"» (ESS ERIC, 2025)

| **Trust cluster** | **Question** | **Ordinal scale** |
|---|---|---|
| *trstlgl*, *trstplc*, *trstsci*, *trstplt* | Please tell me on a score of 0-10 how much you personally trust each of the institutions… | *0 - No trust at all*<br>*10 - Complete trust* |
| **Ideological cluster** | **Question** | **Ordinal scale** |
| *lrscale* | In politics people sometimes talk of "left" and "right". Where would you place yourself on this scale, where 0 means the left and 10 means the right? | *0 - Left*<br>*10 - Right* |
| *gincdif* | To what degree do you agree or disagree with the following statement? 'The government should take measures to reduce differences in income levels.' | *1 - Agree Strongly*<br>*5 - Disagree Strongly* |
| *rlgdgr* | Regardless of whether you belong to a particular religion, how religious would you say you are? | *0 - Not at all religious*<br>*10 - Very religious* |
| *imwbcnt* | Is [Country] made a worse or a better place to live by people coming to live here from other countries? | *0 - Worse place to live*<br>*10 - Better place to live.* |
| *ccrdprs* | To what extent do you feel a personal responsibility to try to reduce climate change? | *0 - Not at all*<br>*10 - A great deal* |

**Table 1**. The selection of variables, their cluster according to Duckitt's RWA/SDO partitioning, the question asked during the interview and their ordinal scale.

Data was acquired by downloading the CSV files for each selected country via the ESS open database. These variables form the basis of the psychometric network analysis, in which each country is represented as a monomodal and undirected network where all nodes represent the same type of entities and the edges represent the positive or negative correlation between them. The nodes represent the attitudinal variables within a country, and a negative edge indicates there is an inverse relationship and vice versa. For example, a negative edge between *trstplt* and *lrscale* suggests that right-wing respondents have low trust in politicians. This offers a granular examination of ideological dynamics within each country, for instance, how trust in institutions correlates with self-reported political orientation.

## Data preprocessing and computing

To preprocess the raw csv files for computing, a subset of data was cleaned in Python using the pandas library. The variable *gincdif* required additional processing: its original 1–5 scale was rescaled to a 0–10 range and reversed, to ensure directional consistency across all variables. In the final coding scheme, higher values consistently represent stronger agreement, greater institutional trust, or a more right-wing position, as established by *lrscale*, where 10 denotes the far-right. Missing values, coded in the ESS codebook as 7, 8, 9, or 77, 88, 99, were replaced with NaN.[5]

Then, the network structures and adjacency matrices were estimated using *EBICglasso* applied to *polychoric correlation* matrices, computed via the *cor_auto* function, to account for the ordinal nature of the survey items. All computations were performed in *R* (4.4.3) using the packages *qgraph*, *bootnet*, and *psych*. Missing values were excluded

[5] The cleaned subsets are available in the GitHub repository: [https://github.com/chiarapicardii/NetworkAnalysis/tree/main/data/csv.]

pairwise during polychoric correlation estimation, which helped to preserve the maximum available data for each variable pair.

To test the null hypothesis of network invariance across countries, we applied a permutation-based Network Comparison Test (NCT; van Borkulo et al., 2023). The NCT null hypothesis states that the network structure of any two compared countries is statistically equivalent. The 11-point survey scales provided sufficient variance to be treated as approximately continuous (Gaussian), providing computational stability across 1,000 permutation iterations. For the NCT procedure too, missing values were removed pairwise, as required by the *NetworkComparisonTest* package.

Finally, the estimated network structures were exported as separate nodes.csv and edges.csv files for editing and visualization in Gephi (Bastian et al., 2009).

## *4.* Validity and Reliability

To guarantee methodological robustness and validity in our model, the focal points were: the authoritativeness of the source, ensuring a reliable basis for the work; an appropriately-sized sample of respondents and a diverse range of European countries; and a structured and grounded theoretical framework minimizing possible bias in the analysis.

**Source and standardization**

The use of the ESS as our data source provides a high level of validity. The ESS, as stated in their documentation[6], is a globally recognised, cross-national project that employs transparent and rigorous methodological standards. According to their Guide to Using Weights and Sample Design (2020) the construction of the sample was a random address-based selection while the data collected was corrected through a post-stratification process based on gender, age, education and geographical region (Kaminska, 2020). This allowed ESS to avoid a priori bias and to limit the influence on the sample to an ex-post normalization, based on each country's demographic composition. Consequently, their rigorous design supports the cross-national ideological comparison of this study. However it is important to highlight how the weights process is based on demographic values, leaving room to residual bias in the trust and ideological variables we focused on, which cannot be fully addressed with post-stratification alone.

**Theoretical construct and variables**

The validity of the Trust and Ideological clusters is rooted in Duckitt's theory (2001), which organizes political attitudes around two independent dimensions, which are reflected in the trust and ideological clusters. By selecting variables that touch upon institutional trust, climate responsibility and social views, both on the redistribution and immigration matters, the study aims to recreate the multilayered picture of modern political issues without relying on just one-dimensional and reductive scale (*lrscale*).

**Reliability**

For several countries, a substantial number of missing values emerged for specific variables, most notably *trstsci* for FR and CZ (>30% of all cases), where this node was absent entirely,

[6] «A key aim of the ESS has always been to implement high quality standards in its methodology and to improve standards in the field of cross-national surveys more generally.» (ESS, 2025)

which ultimately required us to reconsider our initial approach to estimating the network structures and adjacency matrices.

In the initial approach, the Graphical Lasso (GLASSO) was used to compute partial correlation matrices. While widely applied in high-dimensional settings, GLASSO carries two limitations for psychometric network data. First, the L1 regularization penalty shrinks small edge weights to exactly zero, risking the suppression of genuine but weak associations and distorting centrality measures and cluster structure.[7] Second, GLASSO assumes multivariate normality and operates on Pearson correlations, which are not well-suited to ordinal survey scales and can bias the resulting partial correlations (Rhemtulla et al., 2012). Variables with more than 30% missing values were deleted entirely prior to estimation.

We therefore transitioned to *EBICglasso* applied to polychoric correlation matrices (Epskamp & Fried, 2018). Polychoric correlations estimate the latent Pearson correlation between continuous underlying variables from which ordinal responses are assumed to be discretized, making them the appropriate input for Likert-type data (Rhemtulla et al., 2012). EBICglasso selects the regularization parameter λ via the Extended Bayesian Information Criterion, better balancing sparsity against the retention of weak edges. For our data conditions (nine variables, $n > p$), this combination represents current best practice for ordinal psychometric network estimation. This shift resulted in denser networks, with previously isolated nodes (such as *trstsci* in FR and CZ) reintegrating into the network structure.

## 5. Measures and Results

**Measures**

The measures applied were selected to answer a specific analytical question at each level of the network's structure. Together they allow a systematic, multi-level reading of whether Duckitt's (2001) two-factor architecture is present, how internally coherent each cluster is, how the two clusters relate to each other, and which specific nodes are responsible for that relationship.

For the nodes, three centrality measures were applied: weighted degree, which reflects the overall connection strength of each attitudinal variable across the network; eigenvector centrality, which weights those connections by the importance of neighboring nodes and is particularly informative for *lrscale*, since a high value indicates that ideology has fused with the trust cluster rather than remaining structurally independent; and betweenness centrality, which identifies nodes lying on the shortest path between all other pairs and thereby locates the specific variable bridging the RWA and SDO dimensions in each country.

At the network level, modularity was computed using the Louvain method in Gephi (resolution = 1.0, edge weights enabled) to detect whether nodes partition into two communities corresponding to Duckitt's trust and ideology clusters, with three or more communities indicating internal fracture of one dimension; and the mean clustering coefficient (CC) was used as an indicator of overall structural coherence, where a high mean CC signals fusion of the two factors and a low mean CC signals fragmentation. (See Figure 1 in the Appendix for a cross-national comparison of the trust and ideology nodes' clustering coefficients.)

There are two exceptions - France and the Czech Republic - where *trstsci* was excluded from both networks due to missing data, leaving those two networks with eight nodes. All network files are available on the project's GitHub page.

[7] Williams, D. R. (2022). Network Psychometrics: Frequentist Edition (Doctoral dissertation, University of California, Davis).

**Cross-national results**

Across all eight networks, *trstlgl* achieves the highest eigenvector centrality and weighted degree in seven countries (range: 0.98 in CZ to 1.25 in IT), with the single exception of Greece where *trstplc* leads (weighted degree = 1.27, eigenvector = 1.00). The institutional trust triad - *trstlgl*, *trstplc*, *trstplt* - consistently forms the structural core across all countries, as reflected by their dominant centrality values in every network, while *lrscale*, *gincdif*, *rlgdgr*, and *ccrdprs* are consistently more peripheral. At the periphery, *rlgdgr* and *gincdif* return the lowest centrality values overall; Hungary's *rlgdgr* has a weighted degree of 0.18, the lowest of any node in the entire dataset. The most stable positive edges are *trstlgl–trstplc* (mean r = 0.45, range 0.27 in NL to 0.60 in GR) and *trstlgl–trstplt* (mean r = 0.33, range 0.17 in GR to 0.43 in IT). The most consistent negative relationships are *lrscale–gincdif* (negative in all eight countries) and *lrscale–imwbcnt* (negative in seven, all except CZ), confirming that right-leaning self-placement is systematically opposed to redistribution preferences and positive attitudes toward immigrants across nearly every context.

Six of the eight countries produce a two-community modularity solution consistent with the RWA/SDO distinction. However, the inter-cluster relationship varies substantially. The Netherlands (CC = 0.769) produces the closest approximation to Duckitt's model: two distinct clusters loosely coupled through *rlgdgr* (betweenness = 17). France (CC = 0.818) shows RWA-SDO fusion: *lrscale* has eigenvector centrality of 0.653, betweenness of zero, and weighted degree of 0.915; it is fully absorbed into the ideological cluster, not serving as a bridge. Bridging in France falls instead to *gincdif* (betweenness = 8) and *rlgdgr* (betweenness = 8), suggesting that redistribution preferences and religiosity are the structural connectors between the two dimensions, rather than raw ideological self-placement. Italy (CC = 0.791, three communities) presents a more complex fusion: *lrscale* migrates into community 1 alongside *trstlgl*, *trstplt*, and *imwbcnt*, meaning ideological self-placement is structurally absorbed into the trust cluster rather than forming an independent SDO dimension. *Imwbcnt* (betweenness = 11) acts as the primary broker within that merged cluster, while *trstplc* and *trstsci* occupy a second community and *ccrdprs* and *gincdif* fragment into a third, consistent with a polarized context in which outgroup attitudes become entangled with institutional deference, dissolving the boundary between the RWA and SDO dimensions. Finland (CC = 0.663, three communities) represents the opposite pattern: the *gincdif–lrscale* edge (r = −0.316, the strongest negative edge in the dataset) dominates the network but produces fragmentation rather than coherence, splitting the network into three communities and leaving *trstplt* (betweenness = 18, the highest in the dataset) and *ccrdprs* (betweenness = 11) as the bridges between them. Hungary (CC = 0.774, two communities) shows a theoretically distinctive configuration: *trstplt*, *ccrdprs*, *gincdif*, *lrscale, imwbcnt,* and *rlgdgr* form the ideological community, while *trstlgl*, *trstplc*, and *trstsci* remain in the institutional trust community. Trust in politicians has become ideologically coloured rather than institutionally grounded, with *rlgdgr* serving as the top broker (betweenness = 11), consistent with the role of religious conservatism in Hungarian political identity. Great Britain (CC = 0.740, two communities) represents a structurally conventional case: the trust nodes and *rlgdgr* occupy first community, while *lrscale*, *imwbcnt*, *gincdif*, and *ccrdprs* form the second one - a partition that maps closely onto the RWA/SDO distinction. The top broker is *trstplt* (betweenness = 14), suggesting that trust in politicians is the hinge between institutional deference and ideological positioning. The Czech Republic (CC = 0.748, two communities) sits near dealignment: *lrscale* has the highest betweenness in that network at just 6, acting as a weak bridge rather than an organising force, with the SDO dimension leaving minimal structural imprint.

**Network Comparison Test: null hypothesis and result**

The null hypothesis tested by the permutation-based NCT (van Borkulo et al., 2023) here states that the network structures of France and Finland are statistically equivalent and that any observed differences in global strength or individual edge weights are no greater than what would arise by chance under 1,000 random permutations of group membership. France and Finland were selected as the comparison pair because they represent the most structurally contrasting archetypes in the dataset: fusion versus fragmentation, the highest versus lowest clustering coefficient, ideology embedded in the core versus ideology fractured to the periphery.

The null hypothesis is rejected on all primary tests. The results in Table 2 show network invariance test returned $M = 0.181$ ($p = 0.001$), confirming the two structures are statistically distinct. The global strength invariance test returned $S = 0.661$ ($p = 0.007$), with France's global strength at 4.197 and Finland's at 4.858: despite near-identical mean weighted degree ($\approx 0.70$ in both), Finland's attitude variables are significantly more strongly interconnected overall with a higher total edge weight that nonetheless fails to produce higher clustering, characterizing Finland as a network of strong but structurally open connections.

As it can be observed in Tables 3 and 4, ten edges differ significantly between the two countries ($p < 0.05$). The pattern is not random: *lrscale* and *imwbcnt* appear repeatedly, confirming that the structural fault lines between the two countries are precisely the edges that operationalize the relationship between the RWA and SDO dimensions. The centrality invariance test finds that *trstlgl* (strength $C = -0.265$, $p = 0.004$; expected influence $C = -0.277$, $p = 0.004$), *imwbcnt* (expected influence $C = -0.433$, $p = 0.001$), *lrscale* (expected influence $C = -0.334$, $p = 0.012$), *trstplt* (expected influence $C = -0.196$, $p = 0.040$), and *ccrdprs* (expected influence $C = -0.239$, $p = 0.048$) are all significantly more central in Finland than in France. All C-values are negative, indicating that Finland consistently shows higher node influence across every tested variable; its network has stronger, more influential hubs, despite being more fragmented in community structure.

The NCT result directly answers the research question. The null hypothesis of structural equivalence is rejected: national attitudinal networks are not statistically interchangeable. The cross-national differences observed in the descriptive analysis - in clustering, community structure, inter-cluster coupling, and the role of ideology - reflect genuine population-level divergence rather than sampling noise. The two-cluster structure is recoverable across most countries, confirming the general validity of Duckitt's dual-process model. But the relationship between those clusters varies systematically: from fusion, to loose coupling, to fragmentation, to near-dealignment. That variation, statistically confirmed to be real, is the core empirical contribution of this study.

## 6. Conclusion

The central finding of this study is that Duckitt's dual-process model is better understood as a conditional framework than a universal one. The two-factor structure is recoverable where socio-economic and political conditions allow coherent attitudinal clustering around authority and hierarchy. Where those conditions produce tension - as in Finland's welfare-state context, where redistribution is institutionalized as a cross-ideological norm - or where polarization binds the dimensions together - as in France and Italy - the structure shifts in theoretically predictable ways. This context-sensitivity is not a failure of the model but its most informative prediction, and the network approach is well suited to detecting it.

The evidence supporting this conclusion operates at two levels. First, the two-cluster structure Duckitt predicts is broadly present across all eight countries: a stable institutional trust core anchored by *trstlgl* in seven of eight countries (and by *trstplc* in Greece) forms the RWA-related dimension in every national network, and an ideology/outgroup cluster is recoverable as a distinct modularity community in six of eight. This consistency across nations constitutes a general confirmation of the dual-process model: the two attitudinal dimensions are not arbitrary theoretical constructs but empirically recoverable modules in the data.

The second finding is that the relationship between these two clusters is not fixed. What varies across countries is how the RWA and SDO dimensions relate to each other and that variation maps systematically onto national context. In France, ideology (*lrscale*) is fully absorbed into the core alongside the trust variables, producing RWA-SDO fusion: the two dimensions are co-organized and structurally inseparable. In Italy, the same fusion occurs through a different mechanism, *lrscale* is absorbed directly into the trust cluster alongside *trstlgl* and *trstplt*, with immigration attitudes (*imwbcnt*, betweenness = 11) acting as the primary broker within that merged cluster, while climate and redistribution attitudes fragment into a third community. In the Netherlands, the two clusters are present but loosely coupled, connected primarily through religiosity. In Finland, the redistribution-ideology axis is the strongest relationship in the entire dataset (*gincdif–lrscale*, $r = -0.316$), yet its relevance fragments rather than organizes the network: the SDO and RWA dimensions operate independently, connected only through the structurally fragile bridges of trust in politicians (*trstplt*, betweenness = 18) and personal climate responsibility (*ccrdprs*, betweenness = 11). In the Czech Republic the SDO dimension is empirically attenuated, and the network approaches dealignment.

The Network Comparison Test between France and Finland, the two structurally most contrasting cases, confirms that these differences are statistically real rather than sampling artefacts. The null hypothesis of structural equivalence is rejected: $M = 0.181$ ($p = 0.001$) for overall structure, $S = 0.661$ ($p = 0.007$) for global strength. The ten significantly divergent edges cluster precisely around lrscale and *imwbcnt* (the variables that operationalize the interface between the RWA and SDO dimensions) and Finland's nodes are consistently more central in the centrality invariance test, with *imwbcnt* showing the largest differential ($C = -0.433$, $p = 0.001$). The statistical evidence points to genuine cross-national divergence in attitudinal architecture, located exactly where theory predicts it should be.

## 7. Critique

This study partially answers the question it set out to address. It demonstrates that the RWA-SDO structure is recoverable from network data under certain socioeconomic conditions, and that it varies meaningfully across national contexts in ways that are consistent with Duckitt's own theoretical predictions. What it cannot do is confirm the psychological mechanisms behind that variation: because we use proxy variables from a general-purpose survey rather than purpose-built RWA and SDO scales, the study establishes structural plausibility rather than direct measurement. Several further limitations define the boundaries within which the findings should be interpreted.

The most immediate limitation concerns the data collection mode. The ESS responses used in this study were gathered through interviews, which introduces social desirability effects. Respondents may moderate their answers on sensitive topics such as trust in police, attitudes toward immigrants or left-right self-placement in presence of another person. This means some of the observed partial correlations may reflect the interview context as much as the underlying attitudinal structures the RWA-SDO framework seeks to capture. A

comparison between interview-based and self-completion responses within the same dataset would help assess the magnitude of this effect.

The geographic scope also introduces a related constraint. All eight countries are European, which limits the generalizability of the conclusions. Duckitt's model is proposed as a cross-cultural account of ideological organization but testing it exclusively within a single continent with shared historical, institutional, and democratic traditions means the findings cannot speak to whether the same patterns would appear in non-Western or non-democratic contexts. A further comparability issue arises from unequal sample sizes: the largest national sample is Greece (n = 2,799) and the smallest is Great Britain (n = 1,149). While both are sufficient for a nine-node network, the difference in estimation precision means that weak edges are more likely to be reduced by *EBICglasso* in smaller samples, potentially making some structural differences between countries a partial artefact of sample size rather than genuine attitudinal variation.

This study demonstrates structural plausibility rather than direct validation. Because we rely on proxy variables from a general-purpose survey rather than purpose-built RWA and SDO scales, we cannot confirm Duckitt's model - only that its predicted architecture survives contact with independently collected data. A network structure consistent with the RWA/SDO partition could in principle arise from other theoretical frameworks, and using dedicated scales alongside ESS-style variables would allow these structural claims to be directly tested.

**References**


Adorno, T. W., Frenkel-Brunswik, E., Levinson, D. J., & Sanford, R. N. (1950). *The authoritarian personality*. Harper & Brothers.

Altemeyer, B. (1981). *Right-Wing Authoritarianism*. The University of Manitoba Press.

Bastian, M., Heymann, S., & Jacomy, M. (2009). Gephi: an open source software for exploring and manipulating networks. *Proceedings of the International AAAI Conference on Web and Social Media, 3*(1), 361–362. https://doi.org/10.1609/icwsm.v3i1.13937

Duckitt, J. (2001). A dual-process cognitive-motivational theory of ideology and prejudice. *Advances in Experimental Social Psychology, 33*, 44–113. https://doi.org/10.1016/S0065-2601(01)80004-6

Epskamp, S., & Fried, E. I. (2018). A tutorial on regularized partial correlation networks. *Psychological methods, 23*(4), 617–634. https://doi.org/10.1037/met0000167

European Social Survey European Research Infrastructure (ESS ERIC) (2025) *ESS round 10 - 2020. Democracy, Digital social contacts*. Sikt - Norwegian Agency for Shared Services in Education and Research. https://doi.org/10.21338/NSD-ESS10-2020

European Social Survey European Research Infrastructure (ESS ERIC). (2025). *ESS10 integrated file, edition 3.3* [Data set]. Sikt - Norwegian Agency for Shared Services in Education and Research. https://doi.org/10.21338/ess10e03_3

Kaminska, O. (2020). *Guide to using weights and sample design indicators with ESS data*. Institute for Social and Economic Research, University of Essex. https://www.europeansocialsurvey.org/sites/default/files/2023-06/ESS_weighting_data_1_1.pdf

Pratto, F., Çidam, A., Stewart, A. L., Zeineddine, F. B., Aranda, M., Aiello A.. Social Dominance in Context and in Individuals: Contextual Moderation of Robust Effects of Social Dominance Orientation in 15 Languages and 20 Countries: Contextual Moderation of Robust Effects of Social Dominance Orientation in 15 Languages and 20 Countries. *Social Psychological and Personality Science*, *4*(5), 587-599. https://doi.org/10.1177/1948550612473663

Van Borkulo, C. D., van Bork, R., Boschloo, L., Kossakowski, J. J., Tio, P., Schoevers, R. A., Borsboom, D., & Waldorp, L. J. (2023). Comparing network structures on three aspects: A permutation test. *Psychological methods, 2*8(6), 1273–1285. https://doi.org/10.1037/met0000476

Williams, D. (2022). Network Psychometrics: Frequentist Edition. *UC Davis*. https://escholarship.org/uc/item/18r8b26x

## Appendix

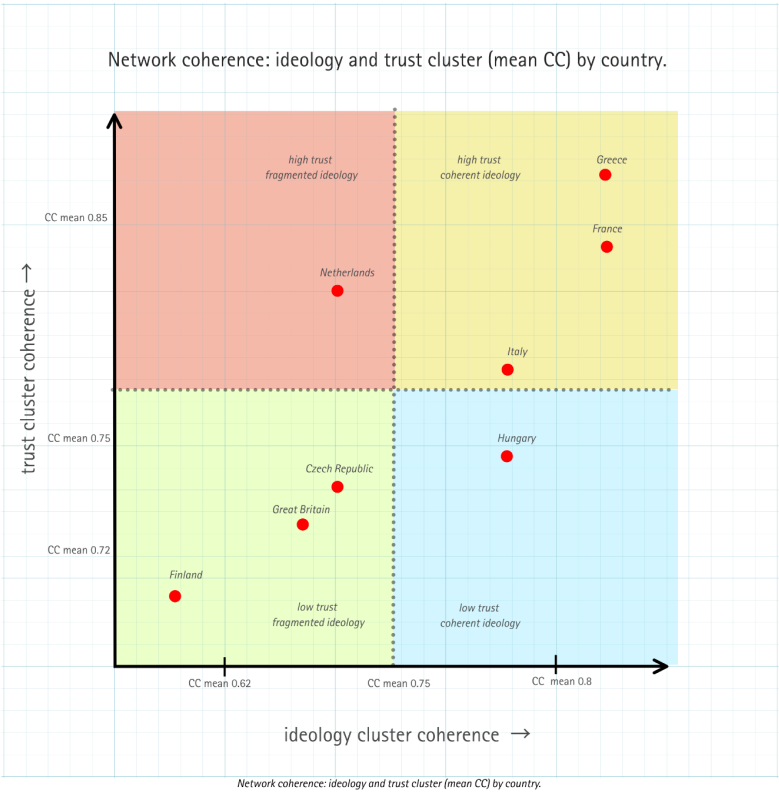


**Fig. 1.** Network coherence: ideology and trust cluster (mean CC) by country.[8]

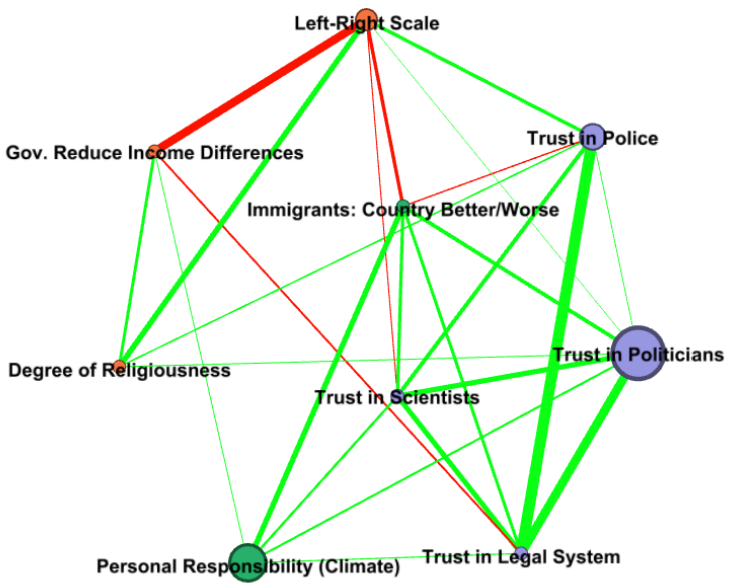


**Figure 2.** Network image of Finland (Node color: Modularity Class/ Node size: Betweenness)

[8] The x-axis reflects the cluster coefficient (CC) mean of the ideology nodes (*lrscale, gincdif, imwbcnt, ccrdprs, rlgdgr*) while the y-axis mirrors the CC mean of the trust cluster *(trstlgl, trstplc, trstplt, trstsci*). Each country's position shows the degree of coherence in the respondents' attitudinal orientations: higher values indicate stronger connections of the nodes, representing a more unified sentiment in the cluster.

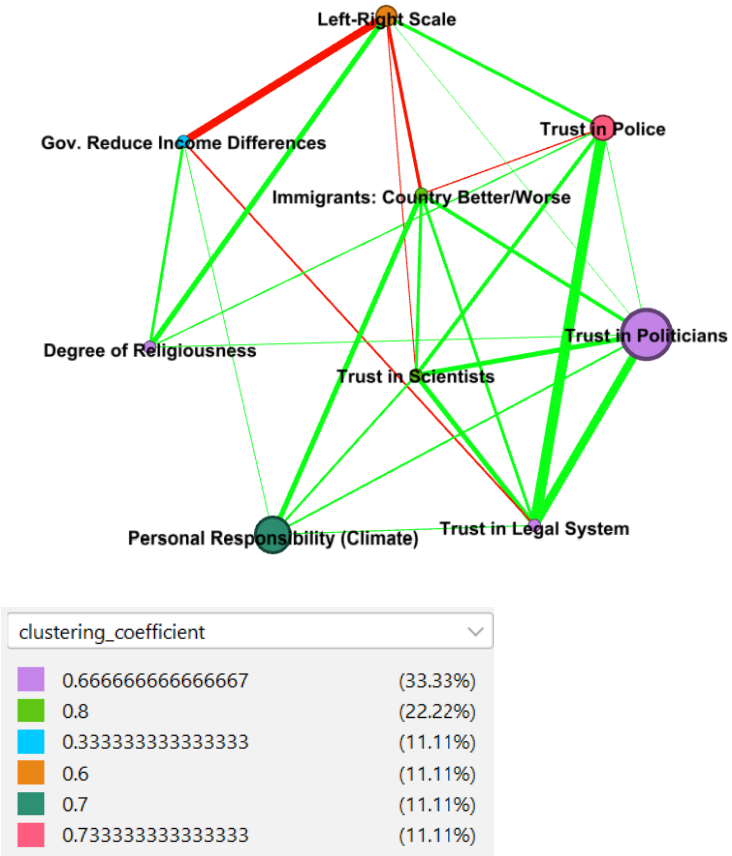


**Figure 3.** Network image of Finland (Node color: Clustering Coefficient/ Node size: Betweenness)

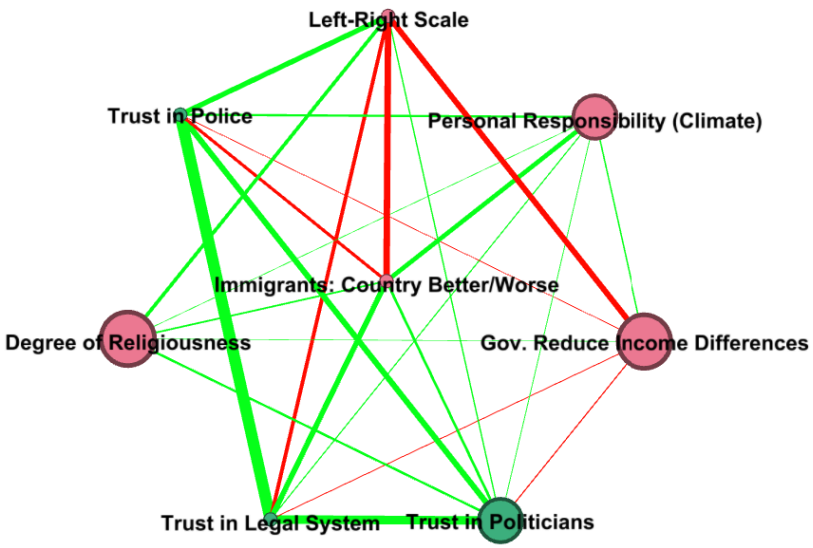


**Figure 4.** Network image of France (Node color: Modularity Class/ Node size: Betweenness)

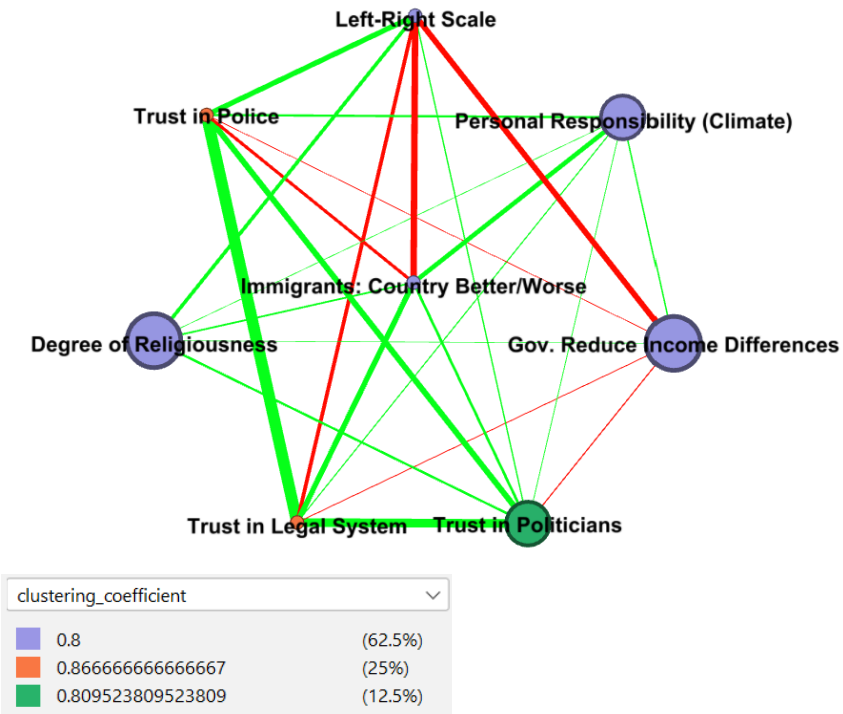


**Figure 5.** Network image of France (Node color: Clustering Coefficient/ Node size: Betweenness)
The rest of the networks are available here: [https://github.com/chiarapicardii/NetworkAnalysis/tree/main/img]

| **Overview of Network Comparison - France vs. Finland** | | | |
|---|---|---|---|
| Test | Statistic | Value | Result |
| Network Invariance | M | 0.181 | p = 0.001 *** |
| Global Strength (France) | S | 4.197 | — |
| Global Strength (Finland) | S | 4.858 | p = 0.007 ** |
| Difference in Global Strength | S | 0.661 | p = 0.007 ** |

Note: *** p < 0.001, ** p < 0.01, * p < 0.05. M = maximum edge difference; S = difference in global strength.

**Table 2.** Overall result of the Network Comparison Test of France and Finland

| **Significant Edge Differences (p < 0.05) - France vs. Finland** | | | |
|---|---|---|---|
| Node 1 | Node 2 | E statistic | p-value |
| trstlgl | lrscale | 0.181 | 0.001 *** |
| trstplt | imwbcnt | 0.142 | 0.001 *** |
| lrscale | imwbcnt | 0.133 | 0.001 *** |
| trstplt | ccrdprs | 0.112 | 0.004 ** |
| gincdif | lrscale | 0.107 | 0.003 ** |
| ccrdprs | imwbcnt | 0.097 | 0.007 ** |
| lrscale | rlgdgr | 0.094 | 0.010 * |
| trstplc | imwbcnt | 0.086 | 0.033 * |
| trstlgl | gincdif | 0.082 | 0.019 * |
| trstlgl | trstplt | 0.052 | 0.048 * |

Note: E statistic = absolute difference in edge weight between France and Finland.

Negative edges and lrscale-related edges account for the majority of significant differences.

**Table 3.** Edge differences of the Network Comparison Test showing significant differences of a p-value of under 0.05

| **Centrality Invariance - France vs. Finland** | | | | |
|---|---|---|---|---|
| Node | Centrality | C statistic | p-value | Direction |
| trstlgl | Strength | -0.265 | 0.004 ** | FI > FR |
| trstlgl | Exp. Influence | -0.277 | 0.004 ** | FI > FR |
| trstplt | Exp. Influence | -0.196 | 0.040 * | FI > FR |
| ccrdprs | Exp. Influence | -0.239 | 0.048 * | FI > FR |
| lrscale | Exp. Influence | -0.334 | 0.012 * | FI > FR |
| imwbcnt | Exp. Influence | -0.433 | 0.001 *** | FI > FR |

Note: Negative C values indicate Finland shows higher centrality.

FI > FR = Finland centrality exceeds France. All significant nodes show higher centrality in Finland.

**Table 4.** Centrality Invariance from the Network Comparison Test